

■ Learning Check

1. An iron nail is brought near a magnet. Which of the following is *not* true?
 - (a) The nail has both a north pole and a south pole.
 - (b) The nail becomes either a north pole or a south pole.
 - (c) The magnet exerts an attractive force on the nail.
 - (d) The nail exerts an attractive force on the magnet.
2. (True or False.) The shape of the magnetic field around a bar magnet is almost exactly the same as the shape of the electric field around a positive charge.
3. Magnetic _____ indicates how far away from true north a compass needle points.

ANSWERS: 1. (b) 2. False 3. declination

■ Learning Check

1. (True or False.) A stationary electric charge produces a magnetic field around it.
2. How does one reverse the polarity (switch the north and south poles) of an electromagnet?
3. A magnetic field exerts a force on an electric charge if the charge is _____.
4. A current can be made to flow in a coil of wire if
 - (a) it is connected to a battery.
 - (b) it is put into motion near a magnet.
 - (c) a magnet is put into motion near it.
 - (d) All of the above.
5. When the coil in a simple electric motor is turned—by a crank, for example—the motor functions as a _____.

ANSWERS: 1. False 2. reverse the current 3. moving 4. (d) 5. generator

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1. A magnetic field will be produced at some point in space if the electric field at that point
 - (a) gets stronger.
 - (b) gets weaker.
 - (c) changes direction.
 - (d) All of the above.
2. (True or False.) A transformer works with AC but not DC.

ANSWERS: 1. (d) 2. True

■ Learning Check

1. What simple thing can you do with a charged object to make it generate an electromagnetic wave?
2. Which of the following is *not* a common use of microwaves?
 - (a) cooking
 - (b) radar
 - (c) medical imaging
 - (d) communication
3. If we see two objects that have different colors, the light waves coming from them must have different _____.
4. Tanning and sunburn are caused by the _____ component of sunlight.
5. (True or False.) Bones show up in x-ray images because they don't absorb x-rays as effectively as muscle and other tissue does.

ANSWERS: 1. make it oscillate 2. (c) 3. frequencies 4. ultraviolet 5. False

■ Learning Check

1. The total radiant power emitted by a blackbody is proportional to its _____ raised to the fourth power.
 2. (True or False.) A glowing blackbody that appears bluish in color is hotter than one that appears reddish.
 3. The blackbody radiation emitted by something as hot as the Sun consists of
 - (a) infrared waves.
 - (b) visible waves.
 - (c) ultraviolet waves.
 - (d) All of the above.
 4. (True or False.) Visible light is the only type of EM wave that living organisms can detect.
- ANSWERS: 1. temperature 2. True 3. (d) 4. False

■ Learning Check

1. Why would a decrease in the amount of ozone in the atmosphere increase the likelihood of people getting sunburn?
2. Of the following gases, which is the most important contributor to Earth's greenhouse effect?
 - (a) water vapor
 - (b) ozone
 - (c) CFCs
 - (d) methane
3. (True or False.) The Hubble Space Telescope and devices that detect nonvisible segments of the EM spectrum are sometimes placed in orbit to overcome the distortion and absorption caused by Earth's atmosphere on EM waves passing through it.

ANSWERS: 1. Increases solar UV 2. (a) 3. True